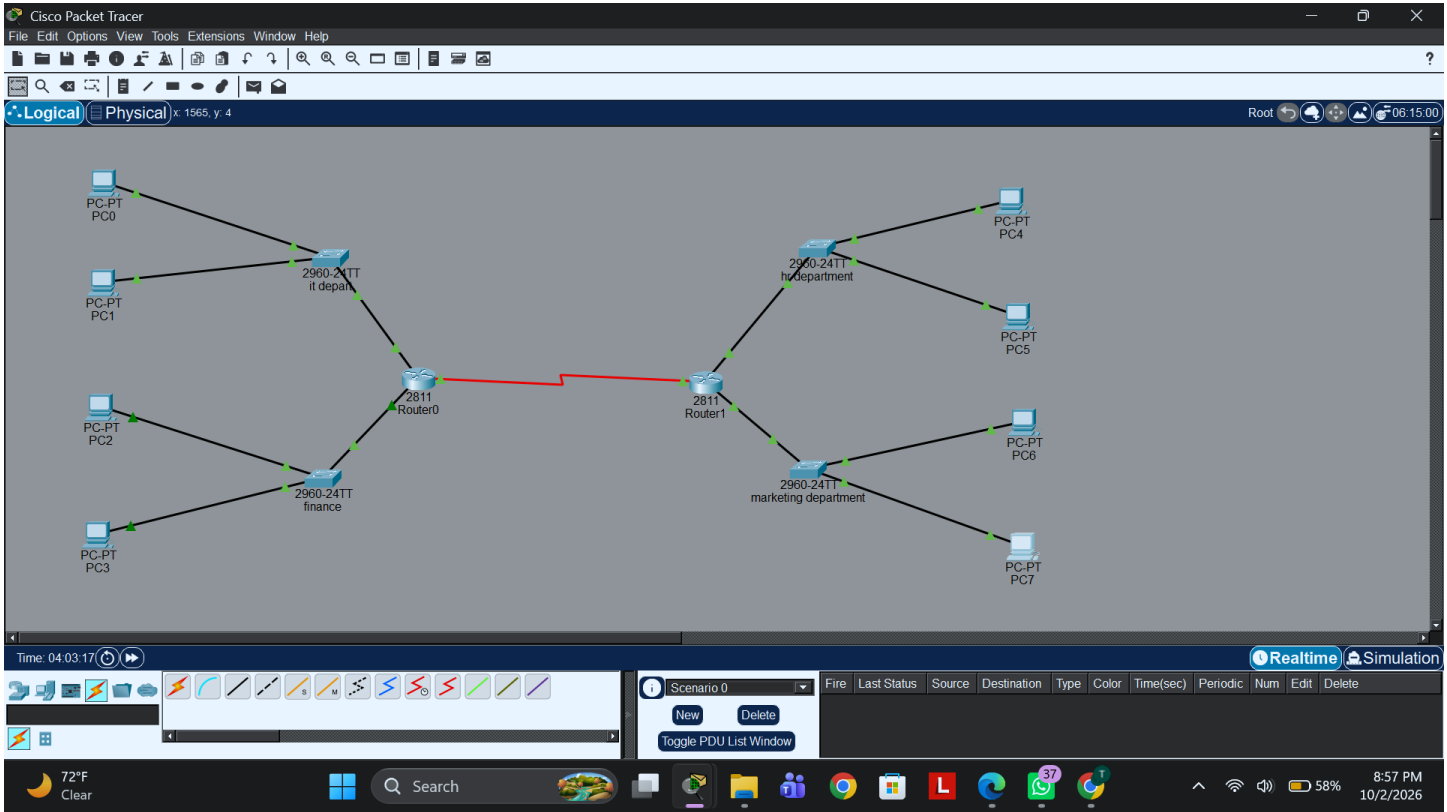


VLSM Multi-Department Network Design

Cisco Packet Tracer Portfolio Documentation — subnetting, IP addressing, gateways, static routing and connectivity verification.

1. Original Network Topology

The original Packet Tracer topology contains four departmental LANs, four switches, eight PCs, two routers and a point-to-point R1–R2 connection.



2. VLSM Calculation — Clearly Explained

The base network is 192.168.50.0/24. VLSM was used to give each department a subnet based on its host requirement. Total addresses = 2ⁿ and usable host addresses = 2ⁿ – 2. The two addresses removed are the network address and broadcast address.

Dept.	Hosts Needed	Host Bits (n)	Total Addresses	Usable Calculation	Usable Hosts	Prefix
IT	120	7	2 ⁷ = 128	128 – 2 = 126	126	/25
Finance	50	6	2 ⁶ = 64	64 – 2 = 62	62	/26
HR	26	5	2 ⁵ = 32	32 – 2 = 30	30	/27
Marketing	5	3	2 ³ = 8	8 – 2 = 6	6	/29

Subnet masks: /25 = 255.255.255.128 | /26 = 255.255.255.192 | /27 = 255.255.255.224 | /29 = 255.255.255.248

3. Complete Subnet Addressing

Dept.	Network	First Host	Last Host	Broadcast	Mask
IT	192.168.50.0/25	192.168.50.1	192.168.50.126	192.168.50.127	255.255.255.128
Finance	192.168.50.128/26	192.168.50.129	192.168.50.190	192.168.50.191	255.255.255.192
HR	192.168.50.192/27	192.168.50.193	192.168.50.222	192.168.50.223	255.255.255.224
Marketing	192.168.50.224/29	192.168.50.225	192.168.50.230	192.168.50.231	255.255.255.248

4. PC IP Addressing and Default Gateways

The portfolio records the addressing plan used for the eight PCs. Each PC receives an address from its department's usable range and uses the router interface on that subnet as its default gateway.

PC	Department	PC IP Address	Subnet Mask	Default Gateway
PC0	IT	192.168.50.2	255.255.255.128	192.168.50.1
PC1	IT	192.168.50.6	255.255.255.128	192.168.50.1
PC2	Finance	192.168.50.136	255.255.255.192	192.168.50.129
PC3	Finance	192.168.50.140	255.255.255.192	192.168.50.129
PC4	HR	192.168.50.195	255.255.255.224	192.168.50.193

PC	Department	PC IP Address	Subnet Mask	Default Gateway
PC5	HR	192.168.50.200	255.255.255.224	192.168.50.193
PC6	Marketing	192.168.50.228	255.255.255.248	192.168.50.225
PC7	Marketing	192.168.50.229	255.255.255.248	192.168.50.225

Addressing rule: network and broadcast addresses cannot be assigned to PCs. For example, Finance uses 192.168.50.129–192.168.50.190 for hosts; .128 is the network address and .191 is the broadcast.

5. Router Interface Addressing

Router	Connection	IP Address	Subnet Mask
R1	IT LAN / Gateway	192.168.50.1	255.255.255.128
R1	Finance LAN / Gateway	192.168.50.129	255.255.255.192
R1	R1–R2 Link	192.168.50.233	255.255.255.252
R2	HR LAN / Gateway	192.168.50.193	255.255.255.224
R2	Marketing LAN / Gateway	192.168.50.225	255.255.255.248
R2	R1–R2 Link	192.168.50.234	255.255.255.252

6. R1–R2 Point-to-Point /30 Subnet

The router link requires only two usable addresses. A /30 provides four total addresses: one network address, two usable host addresses and one broadcast address.

Address Type	Address
Network	192.168.50.232
R1	192.168.50.233
R2	192.168.50.234
Broadcast	192.168.50.235
Subnet Mask	255.255.255.252 (/30)

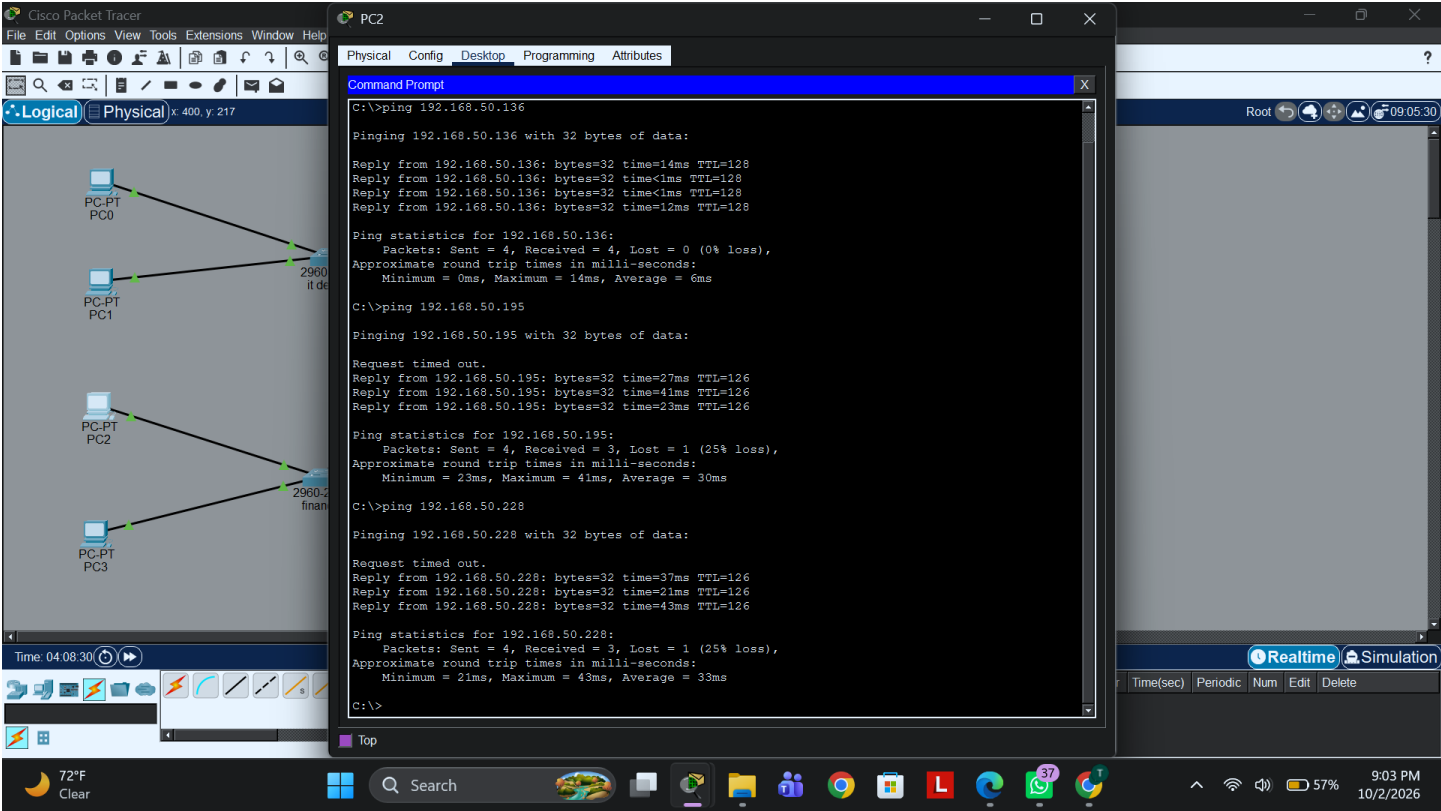
7. Static Routing

Router	Destination	Mask	Next Hop	Purpose
R1	192.168.50.192	255.255.255.224	192.168.50.234	Reach HR
R1	192.168.50.224	255.255.255.248	192.168.50.234	Reach Marketing
R2	192.168.50.0	255.255.255.128	192.168.50.233	Reach IT
R2	192.168.50.128	255.255.255.192	192.168.50.233	Reach Finance

R1 forwards traffic for HR and Marketing to R2, while R2 forwards traffic for IT and Finance to R1. This provides the path required for communication between the four departmental networks.

8. Connectivity Test Evidence

The original command-prompt screenshot below shows real ping tests from the Packet Tracer PC.



9. Ping Results Explained

Destination	Result	What It Shows
192.168.50.136	4 received / 0% loss	Successful communication with a Finance-subnet host.
192.168.50.195	3 received / 25% loss	Three replies received after one initial timeout.
192.168.50.228	3 received / 25% loss	Three replies received after one initial timeout.

The initial timeout can occur while ARP/routing information is being resolved in Packet Tracer. The subsequent replies provide evidence that the destinations were reachable.

10. Portfolio Project Summary

This project demonstrates practical skills in IPv4 addressing, VLSM subnet design, calculation of network/host/broadcast ranges, subnet-mask selection, PC configuration, default gateways, router interfaces, static routing and connectivity verification using Cisco Packet Tracer.

Evidence included: original topology, complete VLSM mathematics, all subnet ranges, PC addressing, router addressing, /30 router link, static routes and ping-test results.